

Fellow Travelers or Worlds Apart?

Posing the Galilean Challenge to Usage-Based Linguistics

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Abstract

Generative Linguistics (GL) and Usage-Based Linguistics (UBL) diverge from one another on how to frame the cognitive science of language. GL seeks to characterize the *knowledge* of language independently of arbitrary instances of its use. UBL, for its part, conceives of the *knowledge* of language and the *use* of language as inextricable, co-mingled during acquisition in ways that make their separation unsatisfactory. This paper argues that this divide between GL and UBL neglects a striking fact: GL's leading expositor, Noam Chomsky, has for decades introduced the foundations of this approach to language with explicit and nearly romantic references to its *use*. The "Galilean Challenge" that motivates a properly conceived science of language, for Chomsky, is to explain how it is that individuals can *select* from an unbounded range of possible expressions an utterance that is *appropriate* to situations but not *caused* by situations. This paper thus asks: Can UBL meet the Galilean Challenge? It answers this through an assessment of three possible means by which the UBList may meet the Challenge. It is found that UBL is unable to meet the Challenge without diluting its core tenets, turning principally on fundamental incompatibilities between itself and GL. A discussion of these findings suggests these results are regrettable and proposes an alliance between GLists and UBLists by showing that the Galilean Challenge's depiction of ordinary human language use successfully pre-empts concerns now raised by some factions in UBL over the role of Large Language Models in linguistics.

Keywords: Creative Aspect of Language Use; Free Linguistic Behavior; Galilean Challenge; Generative Linguistics; Third-Way Linguistics; Usage-Based Linguistics

1 Introduction

An oddity has accompanied discussions about generative linguistics (GL) and its broadest competing approach, usage-based linguistics (UBL). The two approaches to the study of human language, in their most basic formulations, diverge on how to frame the study of language's basis in the human mind. Generative linguists (GLists) indeed embed assumptions in their inquiries foreign to usage-based linguists (UBLists), separating *knowledge* of language from its *use*. UBLists reject this separation, conceiving of linguistic structure as emerging *from* use (Newmeyer, 2021, 291-292). The upshot, discussions normally indicate, is that GLists *delimit* the scope of linguistic inquiry to a fixed “competence” at the expense of linguistic *behavior*, viewed by UBLists as critical to the development and change of this competence.¹

Yet, this is odd given that Noam Chomsky, the leading progenitor of GL, frequently introduces the foundations of this approach to language with explicit and nearly romantic references to its *use*. This has been remarkably consistent since at least the mid-1960s, with the first chapter of Chomsky's 1966 *Cartesian Linguistics* devoted to the “creative aspect of language use,” where he cites with approval Descartes' view that language is “available for the free expression of thought or for appropriate response” and “that there is no man so imperfect as not to use language for the expression of his thoughts...” (Chomsky, 1966, 5). Chomsky has routinely affirmed that language can be “freely” recruited:

It is because of this freedom from stimulus control that language can serve as an instrument of thought and self-expression, as it does not only for the exceptionally gifted and talented, but also, in fact, for every normal human (Chomsky, 2006, 11).

Over decades, Chomsky remained insistent:

We still have no way to come to terms with what appears to be a fact, even an obvious fact: Our actions are free and undetermined, in that we need not do what we are “incited and inclined” to do; and if we do what we are incited and inclined to do, an element of free choice nevertheless enters (Chomsky, 2011, 241).

So important is this “free” linguistic behavior to Chomsky that explaining it is the “Galilean Challenge”:

In the *Discourse on Method*, Part V, Descartes argued that the creative use of language marked the distinction between human beings and other animals, and between human beings and machines. A machine may be impelled to act in a certain way, but it cannot be inclined; with human beings, it is often the reverse. Explaining why this is so, is the Galilean challenge (Chomsky, 2017, 1).

For Chomsky, the characteristics of “creative” language use – its unbounded scope, freedom from detectable stimuli, and appropriateness to situations - converge to form the Galilean Challenge.

It is odd that there are vanishingly few, if any, direct engagements with Chomsky's formulation of the Galilean Challenge by his usage-based critics. It is questionable that these views – which, at least at first

¹ This is a simplification for illustrative purposes, as some proponents of “third-way linguistics” argue that the strict dichotomy between *innate and therefore unlearned knowledge* and *learnable and therefore not innate knowledge* is false (see, e.g., Rastelli, 2025, 4-6).

glance, suggest sympathies with one another's exaltation of the *ordinary uses* of human language apart from the study of "formal" notions of grammar critiqued by UBLists – should remain isolated.

This paper begins filling this gap by posing the Galilean Challenge to UBL. This is conceived as the challenge of explaining how it could be that human linguistic behavior appears to be free of identifiable stimuli, frequently novel, yet appropriate to the circumstances in which it arises. The driving question of this paper is thus: Can usage-based linguistics meet the Galilean Challenge?

This paper's motivation is conciliatory: these two approaches appear simultaneously to be fellow travelers yet worlds apart in their respective extolling of the importance of ordinary language *use*. Whether they are compatible or complementary in this respect is uncertain, but presumably desirable. To this end, the foundations of both GL and UBL are explicated, the former through the lens of the Galilean Challenge. With these foundations, three plausible propositions by which UBLists could meet the Challenge are assessed: (1) GL and UBL are fundamentally compatible in their respective dealings with the Galilean Challenge; (2) UBL can meet the Galilean Challenge on its own terms, GL notwithstanding; and (3) The Galilean Challenge is ill-formulated, shown with UBL-specific resources.

It is found that UBL is unable to meet the Galilean Challenge without diluting its core tenets. Specifically, UBL and GL are found to be fundamentally incompatible in their respective attempts to meet the Challenge, with the adjustments required of UBL too severe to maintain the integrity of the approach's core tenets. Largely in virtue of its fundamental incompatibility with GL, UBL is unable to meet the Challenge on its own, UBL-specific terms. Finally, UBL is unable to show, using its own approach-specific resources – particularly observations about the frequency of novel constructions in actual language use – that the Challenge is ill-formulated.

The word "free" will be used in scare quotes throughout this article. This is merely to convey that the article intends to avoid broader matters related to free will and instead draws directly from the Galilean Challenge as it is formulated within GL.

The paper begins with an explication of GL by breaking down how it meets the Galilean Challenge, particularly through Chomsky's work, focusing on matters including the competence-performance distinction and the modularity and domain-specificity of the postulated language faculty. It explicates the foundations of UBL thereafter in parallel fashion. Then, the three plausible usage-based responses to the Galilean Challenge are assessed, with particular focus on the first – whether UBL and GL are fundamentally compatible. The paper closes with a discussion of the findings, including the limitations of the arguments herein. Against the finding of fundamental incompatibilities, it is suggested that the Galilean Challenge's depiction of "free" linguistic behavior successfully pre-empts concerns that some UBLists currently express on the role of Large Language Models in linguistics. This is proposed as the basis for an alliance between GL and UBL.

2 Foundations of Generative Linguistics

For GLists, the fact that most basically motivates the study of language is the human ability to produce and understand an unlimited number and variety of sentences "with no feeling of difficulty or strangeness" (Chomsky, 2006, 11). In his 1964 *Current Issues in Linguistic Theory*, Chomsky writes:

The central fact to which any significant linguistic theory must address itself is this: a mature speaker can produce a new sentence of his language on the appropriate occasion, and other speakers can understand it immediately, though it is equally new to them...In fact, the realization that this “creative” aspect of language is its essential characteristic can be traced back at least to the seventeenth century...In any event, whatever the antiquity of this insight may be, it is clear that a theory of language that neglects this “creative” aspect of language is of only marginal interest (Chomsky, 1964, 7-8).

This remark describes two problems, though packaged into one. One is the problem of how human beings *use* language to express new thoughts appropriately (linguistic behavior); the other is the *capacity* for infinite generativity from finite means that subserves this use. The “theory of language” that takes seriously language’s “creative” aspect is a theory of the cognitive capacity of language – but not its use.

Chomsky’s allusion to seventeenth century thought was elaborated at length in his 1966 *Cartesian Linguistics*. There, he begins with a summary of the seventeenth century Cartesian conception of human language as an instrument of “free” expression of thought over an unbounded range:

In short, then, man has a species-specific capacity, a unique type of intellectual organization which cannot be attributed to peripheral organs or related to general intelligence and which manifests itself in what we may refer to as the ‘creative aspect’ of ordinary language use – its property being both unbounded in scope and stimulus-free. Thus Descartes maintains that language is available for the free expression of thought or for appropriate response in any new context and is undetermined by any fixed association of utterances to external stimuli or physiological states (identifiable in any noncircular fashion) (Chomsky, 1966, 4-5).

Humans possess a species-specific ability – common to all its members without varying in any discernable measure, save for developmental disorders and the like – to express thoughts over an unbounded range of utterances without fixed reliance on identifiable stimuli. These expressions are routinely given an interpretation of appropriateness by those who hear them. Language use is therefore “free,” in the dual sense that expressions can be *constructed* in novel ways and that these expressions are *selected* by individuals for use without being fixed to identifiable stimuli.

The problem of this “creative aspect of language use” can be put as follows: the human use of language is *appropriate* to situations but not *caused* by situations. Individuals are not *fixed* to a situation yet, within this situation, they select from an infinite class of possible linguistic expressions a mere *subset* of them which are then interpreted as appropriate *to* the situation by others who hear them; uncaused yet appropriate. Although one can recount the specifics of a given set of circumstances within which language is used – the individuals present, the objects within the space, the weather at the time of speaking, etc. – there appears to be no set of utterances that are fixed to identifiable stimuli. Moreover, attempts to identify a particular cause for an individual’s utterance tend, on this view, to undermine scientific explanation by reducing the content of the notion *cause* to a post-hoc attribution. Chomsky’s critique of Skinner’s (1957) notion of stimulus control shows the seeds of this reasoning:

We identify the stimulus when we hear the response...the talk of stimulus control simply disguises a complete retreat to mentalistic psychology. We cannot predict verbal behavior in terms of the stimuli in the speaker’s environment, since we do not know what the current stimuli are until he responds (Chomsky, 1959, 52).

This post-hoc attribution of stimulus control means that “the word *stimulus* has lost all objectivity in this usage. Stimuli are no longer part of the outside physical world; they are driven back into the organism” (Chomsky, 1959, 52).

In any event, the notion of *cause*, for Chomsky (e.g., 2014, 138-139), frequently amounts to a mere re-statement of the specifics of the situation. Moreover, it may provide only a *rational*, but not scientific, explanation – an individual’s *reasons* for acting whose relationship with prior states has not been systematically established – rather than a cause that systematically links an identifiable stimulus with an utterance (see, McGilvray, 2001, 13-18; McGilvray, 2016, 87-88). Indeed, he has questioned whether the notion of *causality* in its application to human behavior is adequately specified:

Again, consciousness makes the difference, along with “causation of behavior,” a notion with its own nontrivial problems. We have little reason to believe that normal behavior *is* caused, at least in any known sense of the term, nor would a methodological naturalist dogmatically assume otherwise (Chomsky, 1994, 199).

McGilvray echoes Chomsky’s allegation of ‘dogmatic’ thought by arguing that the existence of prior internal states and existing external conditions are often merely assumed to constitute *causes*, in some meaningful sense of the term: although one may be ‘incited and inclined’ to think or speak a certain way – with prior thoughts and external conditions providing *reasons* to do so² – “neither prior nor simultaneous conditions constitute causes” (McGilvray, 2016, 88). That they are taken to constitute causes is, as Potter et al. (2026, 2-6) observe generally, frequently an unquestioned starting assumption in free will debates.

It suffices to note the unlimited diversity of situations in which individuals may use language – recruitable, as it were, for any purpose across time and in any context, all while retaining the character of intentional behavior (see, Clayton, 2015). This is observed alongside the apparent ability to *not* use language despite the availability of what may be, in other contexts, deemed *causes*. Language use is “free,” again, in a dual sense: the undetermined *association of words* in the construction of expressions, on the one hand, and the *selection* of novel expressions *without* reliance on identifiable stimuli in the local environment, on the other. It is the ability “to respond appropriately to the input while still being “detached” from that input” (Boeckx, 2010, 36). When Chomsky claims that language is “available for the free expression of thought precisely because it is not tied directly to external stimuli” (Chomsky, 1975, 302), this is his point.

Appeals to the internal physiological states of individuals are non-starters for resolving this problem. It is not that there are no such bases. It is, rather, that language appears to serve as the means by which the “free” deployment of intellectual resources can be undertaken, configured appropriately to produce a “meeting of minds” (Collins, 2006, 489).

2.1 Justifying the Competence-Performance Distinction

² Invoking *reasons-as-causes* reduces the task of explanation to post-hoc attribution, as *stimulus control* did for Skinner; we do not attribute a reason-as-cause until the individual uses language, and only part of an interpretation of the event. More problematically, the search for causal accounts of human behavior appears to invoke reasons-as-causes only after the failure to identify a “brute cause which is sufficient to explain the behavior” (McDonough, 1994, 128).

To be sure, things have changed since the seventeenth century. The establishment of general computability theory in the early twentieth century, and the recognition that “[i]t is possible to invent a single machine which can be used to compute any computable sequence” (Turing, 1937, 241), allowed the Cartesian problem to be reformulated, splitting a linguistic *competence* capable of characterizing infinite structured expressions from linguistic *performance* (Watumull and Chomsky, 2020, 4). Universal Grammar (UG) was taken to be the theory of “a finite object that characterizes an infinite array of ‘free expressions,’ each a mental structure with a certain form and meaning” (Chomsky, 2007, 45). Yet, UG targeted only the “infinite scope of finite means” (Chomsky, 2007, 46); the capacity for infinite generativity. The postulation of an innate “language faculty” provides only the *basis* for “free” linguistic behavior, but not behavior itself.

On explaining “free” linguistic behavior, Chomsky (e.g., 2006, 11) has been consistently pessimistic. The “creative” aspect of language use is taken to be scientifically intractable in its entirety, at least for now. GLists therefore designate language *use* beyond the scope of foreseeable scientific explanation (Asoulin, 2013; Chomsky, 2006, 86). A distinction between *competence* and *performance* is invoked to limit the target of inquiry to the tractable problem of the knowledge of language.³

2.2 The Domain-Specificity and Modularity of the Language Faculty

This depiction of “creative” linguistic behavior leads GLists to a particular conception of cognitive architecture. Because the GList holds that linguistic behavior is scientifically intractable, the study of human language turns inward; an *internalist* science of language in which linguistic competence is studied (see, McGilvray, 2001; Asoulin, 2013). As a result, GL is concerned with studying the cognitive architecture necessary to support the *possibility* of “free” linguistic behavior.

Unboundedness is, most straightforwardly, accounted for by postulating an innate, domain-specific generative language faculty that can yield an infinite array of structured linguistic expressions from finite elements. This was made possible, as noted, by the establishment of general computability theory.

Stimulus-freedom and appropriateness are separate problems, as the cognitive basis for language must regulate the production of structured expressions *without* reliance on identifiable stimuli in the local environment while nevertheless being appropriate to the situation at hand. That is, this cognitive basis must afford an operational autonomy. Therefore, beyond enabling unboundedness, this cognitive basis must fulfill two responsibilities: (1) the generation of structured expressions independent of identifiable stimuli in the local environment (self-induced production); and (2) interfacing with separate but relevant cognitive systems in the mind/brain whose “inputs” are necessary to “fit” the expression(s) to the situation.⁴ On McGilvray’s account, a *generative* and *modular* language faculty meets both conditions:

The possibility of stimulus freedom in language use can be seen to result from a modular language faculty. Not just any kind of modularity will do: we need a faculty that utilizes its own

³ This is not the sole route to the competence-performance distinction. Perhaps more frequently, GLists justify this methodological constraint by arguing that the variability of human behavior *confounds* explanatory efforts (see, e.g., Volenec and Reiss, 2025).

⁴ Remember, all three conditions – stimulus-freedom, unboundedness, and appropriateness – must be met by one’s postulated cognitive architecture. Although linguists typically focus on matters subsumed by unboundedness, it is not enough that one establish the conditions for novelty to arise; it must also, to meet the Galilean Challenge, enable stimulus-free and appropriate language use.

resources and with internal prompting produces (in apparent isolation) through its own algorithms items in the form of linguistic expressions that are unique to it and yet that “interface” with relevant other internal biological systems (McGilvray, 2001, 12).

The kind of modularity referenced by McGilvray is taken up by Chomsky elsewhere, characterizing the language faculty as “a subcomponent of a more complex system with enough internal integrity to be studied independently as part of a broader investigation of its interactions with other such systems in the life of the organism...” (Chomsky, 2013, 35). This modularity entails that the language faculty comes equipped with its own “conceptual resources” and can make these resources “legible” to other systems in the mind (McGilvray, 2005, 218).

To be sure, views on the modularity of the language faculty in GL are varied and often underspecified. Modularity itself is reducible to (at least) two notions including “the idea that language is a distinct module of the mind; on the other hand, the idea that this module is characterized by modular structure in itself” (Pleyer & Hartmann, 2019, 4). These authors elsewhere suggest GLists and UBLists are no longer “asking how the modular ‘language organ’ came into existence but instead are taking a ‘mosaic’ view of the evolution of the different components that make up the language ready brain” (Hartmann & Pleyer, 2020, 142). One should be careful about moving too far in this holistic direction, however. Mendívil-Giró correctly points out that UG – taken to be the study of the “*initial* formulation of the properties of [linguistic] entities” (Mendívil-Giró, 2018, 866) - assumes the existence of a language faculty *by definition*, with GLists targeting the *specifically linguistic* biological resources that yield generativity (Mendívil-Giró, 2018, 862-866). That specifically linguistic entities are reducible to non-linguistic entities is a trivial fact about such an investigation (Mendívil-Giró, 2018, 864-866).

A satisfying answer to the extent to which the language faculty in GL is modular will not be provided here. For our purposes, the exercise of the linguistic capacity independently of identifiable stimuli entails that this capacity is self-prompting, isolable to some extent from other cognitive systems, while capable of interfacing with them. Our claim is minimal: *some specifically linguistic resources are autonomous*.⁵

3 Foundations of Usage-Based Linguistics

Consider the questions that Chomsky (1986, 3) considers central in the study of language:

- What constitutes knowledge of language?
- How is knowledge of language acquired?
- How is knowledge of language put to use?

As Newmeyer (2021, 291) argues, these questions *implicitly* hypothesize that knowledge of language can be characterized separately from its use. The UBList diverges from the GList on this most basic matter.

For the UBList, knowledge of language and use of language are mutually reinforcing and *cannot* be strictly distinguished in its study. The usage-based approach to linguistics, in its broadest formulation,

⁵ Newmeyer argues that there may have been a slight ‘weakening’ of the autonomy of syntax in GL, particularly Chomsky’s work, as a possible “semantic creep” that intrudes upon the “syntactic module” (Newmeyer, 2021, 293). We will not review this further, other than to note that some modularity would nevertheless be retained.

holds that *grammar is the cognitive organization of an individual's experience with language* (Newmeyer, 2021, 288). For UBLists, *structure in language emerges from use* (Bybee, 2023, 13). Knowledge of language for the UBList is *constructed* through use; experience underlies cognitive organization of grammar (Newmeyer, 2021, 291). UBLists therefore reject the implicit hypothesis of GL that knowledge and use of language can be characterized separately.

In this way, language is a “complex adaptive system” (Bybee, 2023, 13), with its cognitive organization grounded in the *social context* of its use (Bybee, 2023, 12). This approach therefore places significant emphasis on the role of *communication* in the cognitive representation of language, as it is through interaction with others that particular expressions are used to convey meanings. The “most important property of constructions,” Bybee writes, “is that they directly associate meaning to form,” in this way allowing constructions to be “conventionalized” such that they *can* convey meanings (Bybee, 2023, 17). Thus, all UBLists take the *frequency* of use of specific constructions to be of central importance; frequency shapes changes in language given that its use is *how* certain forms are grammaticalized at all (Newmeyer, 2021, 289).

As Tomasello notes, the core tenets of the usage-based approach are: *meaning is use* and *structure emerges from use* (Tomasello, 2015, 89). The UBList's focus on the social context of language use owes to a dissatisfaction with purely “formal” notions of grammar, making communicative processes the locus of inquiry into the acquisition of language by children (Tomasello, 2015, 89-90). The important point is that UBLists share an aversion to the abstractly specified, formal notions of grammar in GL and study linguistic structure first and foremost as a malleable product of *use* grounded in social context.

Importantly, for UBL, meaning derives from social context; meaning is (and must be) derived from shared activity, such that alternative meanings are pruned away during interactions. In Tomasello's example, pointing at a toy during clean-up by a participant informs an infant's joint attentional frames. The gesture communicates a meaning, and the meaning is bound to the context (Tomasello, 2015, 90-91). Language use thus has a *function*: the expression of meaning. This meaning, being bound up in experience, makes the *social context* of language use critical to the cognitive bases of language (Newmeyer, 2021, 289). UBL can be glossed as the study of “the ways in which cognitive, social, and interactive pressures shape the linguistic system and how these factors enable humans to use language as fluently as they do” (Bunzeck et al., 2026, 35).

3.1 The Domain-Generality and Non-Modularity of Language

Working backwards from language's function, UBL seeks the optimal cognitive architecture capable of yielding it. UBLists therefore posit *domain-general* cognitive mechanisms: “linguistic structure arises from the general properties of human cognition as manifest in communication...domain-general abilities” (Bybee, 2023, 12-13). Details on how to characterize these domain-general capacities vary among UBLists, as it does for GLists with the domain-specific language faculty. For example, Tomasello promotes two such domain-general capacities: (1) intention-reading, and (2) pattern-finding. Respectively, these capacities allow children to identify the goals or intentions of mature speakers when they speak for social ends and enables children to “go productively beyond the individual utterances they hear...to create abstract linguistic schemas or constructions” (Tomasello, 2015, 90) (the latter addressing the capacity for generativity).

UBLists are generally suspicious of modularity (Newmeyer, 2021, 289). Although the approach is not steadfastly opposed to modularity, it makes sense that it would harbor such suspicion: if knowledge and use of language are inextricable, and the use of language in communicative contexts is ideally supported by domain-general capacities that subserve a range of behaviors, then it makes little sense to posit that language's cognitive basis is segmented off from other cognitive systems; isolated and in possession of its own, peculiar cognitive structure.

3.2 The Uncertain Philosophical Basis of UBL

It is difficult to find a well-specified, unifying philosophical account of UBL. Ibbotson (2013, 8-9) questions whether such an agreed upon philosophy can be identified as UBL's basis, arguing that a plausible candidate might be *non-essentialist* approaches to language that reject a focus on the relationship between *meaning* and *truth* in favor of the relationship between *meaning* and *use*.

In its absence, it is worthwhile to identify the positions that either all or most UBLists identify with the approach. Newmeyer (2021, 289) lists the following:

- *All* UBLists emphasize frequency of use.
- *All* UBLists *reject* the autonomy of syntax, as they directly link grammatical structure with meaning.
- *Most* UBLists *question* the modularity of language (its independence from other cognitive systems).
- *Most* UBLists *downplay* or *reject* the competence-performance distinction, as there is no *a priori* boundary between semantics and pragmatics.

On all points, UBLists differ significantly with the foundational tenets of GL.

4 Posing the Galilean Challenge to Usage-Based Linguistics

For the UBList, the problem of “free” human language use as identified by Chomsky does not appear to arise within the relevant literature. Although UBL is situated in opposition to the generative depiction of and explanation for the “creative” aspect of human language use, *creative* typically stands-in for *unboundedness* in these critiques. That is, when the generative position on language's “creative” aspect is critiqued by a UBList, it is the characterization of the cognitive basis for linguistic generativity and explanation for this ability that is in question, not the broader notion of stimulus-free, unbounded, yet appropriate language use – the Galilean Challenge.

An open question, then, is: Can UBL meet the Galilean Challenge?

Below, we answer this question by exploring three plausible means by which UBL might attempt to meet the Challenge.

5 Are the Generative and Usage-Based Approaches Fundamentally Compatible?

The UBList may argue that GL and UBL are fundamentally compatible with one another in their respective efforts to meet the Galilean Challenge, granting some mutual adjustments that do not threaten the core tenets of either.

To be sure, commentaries on the possible unification of GL and UBL have had mixed reception. Newmeyer, for one, summarizes a sentiment likely held by one segment of the linguistics community by noting that the “principal differences” between the two “are enormous” while holding out for the possibility of “some ultimate rapprochement” (Newmeyer, 2021, 300). In contrast, some scholars have erred towards such a rapprochement in recent years under the moniker of “third-way linguistics,” an approach that conceives of innate grammar and statistical learning as two sides of the same coin (e.g., Rastelli, 2025). Some fellow travelers of GL indeed argue that theories of (some aspects of) linguistic behavior can be conceived as productive complements to GL itself. Consider Rey’s comment:

All hands agree that language is used sometimes for communication, which is likely to be an issue of performance, involving human purposes and general intelligence. Although Chomsky sometimes seems to suggest that the only serious linguistic theories are of underlying grammatical competence, there is absolutely no reason his core theory need exclude further interesting theories of such processes, in particular of speech pragmatics and social and historical patterns in language use, any more than theories of physics exclude theories of engineering or meteorology (Rey, 2020, 153).

Rey is suggesting that there could be a reconciliation of theories of performance with a theory of grammatical competence.⁶

That said, this would likely be met with dissatisfaction by UBLists, as this *retains* a competence-performance distinction derived from the assumption that knowledge of language *can* be studied independent of performance factors. The UBList sees language as having a communicative function, and the exercise of this communicative function *cannot be decoupled* from a human child’s acquired (*constructed*) knowledge of language. Likely unwittingly, Rey’s position, if adopted, would make UBL ancillary to a “core” generative theory, precisely the hierarchy of analysis that UBL deems problematic.

Still, both the UBList and the GList begin their inquiries with observations about ordinary human language *use*. UBLists develop an account of the cognitive architecture of language by positing mechanisms that are optimal for the derivation of meaning from social contexts and the emergence of structure from those contexts in which language is used. UBLists posit that this entails domain-general cognitive mechanisms – like intention-reading – and reject formal notions of grammar that can be characterized independently of language use. The object of inquiry for the UBList therefore includes the broader social context in which the exercise of this capacity is undertaken, taking the context to be inseparable (on any satisfactory account) from its use in the expression of meaning. Observations about the *ways in which language is used* are the starting points from which the UBList develops an account of language acquisition and its cognitive architecture.

⁶ It is worth noting that Rey (2020, 39, 387) is among those few GLists directly critical of Chomsky’s depiction of language use as “free,” in our sense. His position on GL’s compatibility with UBL is therefore natural, though the barriers for unification are lower.

GLists, for their part, *also* begin from observations about language use. The crucial divergence from the UBList is in the characterization of ordinary language use as “free,” in the sense explicated above, and the philosophical stances motivated by this characterization that bear on its study. The observations of language use embedded in the Galilean Challenge can be thought of as a starting point for inquiry that informs a specifically rationalist approach to scientific explanation that *excludes* language *use* from its purview, instead studying linguistic competence (McGilvray, 2016, 92). Linguistic competence is, in this way, *idealized* to cut through intractable performance factors.

From the same level of analysis, then, different points of focus with respect to the phenomenon in question inform significantly different research tracks: the UBList sees language use as critical to a theory of language acquisition and cognitive architecture; the GList sees language use as scientifically intractable and, therefore, segmented off from inquiry, at least temporarily.

5.1 Loci of the Divide

To determine whether these approaches can be made compatible, two points must be addressed:

- (a) the UBList does not traditionally depict human language use according to its stimulus-freedom, unboundedness, and appropriateness;
- (b) the UBList rejects the philosophical assumptions that lend to the GList’s propensity toward idealization of the language capacity.

5.1.1 Acquiring the Capacity for “Free” Language Use

On (a), were the UBList to accept the Galilean Challenge’s depiction, they must then justify their claims that language possesses an *intrinsically communicative function* and that the study of language cannot be decoupled from its *social context*. For, if the GList’s depiction is accepted, then the behavior under study does not have a pre-defined function such as communication owing to its freedom from identifiable stimuli (that is, neither the stimuli to which language use often attends nor the cognitive structures that language use arises from determines the indefinitely many ways in which it is used).⁷ The UBList would seemingly have to make peace with the methodological constraints invoked by the GList to accommodate the Galilean Challenge, including a fairly rigid competence-performance distinction. Additionally, the cognitive architecture postulated on the back of this competence-performance distinction – a domain-specific and modular language faculty – must either be accepted by the UBList, or they must justify an alternative postulation.

It is unclear whether the UBList can make these accommodations without abandoning the core tenets of their approach. If the UBList argues that the competence-performance distinction can be watered down, then they must similarly argue that “free” language use is in fact tractable. This position seems to be a non-starter, however. The Galilean Challenge indicates that there is no set of stimuli (identifiable independent of arbitrary instances of language use) on which a science of ordinary linguistic behavior can

⁷ A familiar debate within linguistics is whether language is primarily a tool for *communication* or primarily a tool for *thought*. Critics of GL tend to argue the former (e.g., Fedorenko et al., 2024) whereas GLists tend to argue the latter (e.g., Watumull, 2024). However, Chomsky and GLists will often invoke “free” or “creative” linguistic behavior in our sense, but it is difficult to find a direct usage-based response to these invocations. Often, Chomsky’s remarks on “creativity” are quoted in this context without further investigation.

be constructed. Language use is *neither* determined (stimulus-free and unbounded) *nor* random (appropriate to circumstances). Any cognitive architecture postulated by the linguist to account for this behavior would *reduce* to the mechanisms that *enable* the behavior, rather than explain the behavior itself. Thus, a fairly rigid competence-performance distinction seems necessary.

Beyond the competence-performance distinction, the UBList may argue that language's domain-general and non-modular cognitive bases are sufficient to yield this behavior, thereby rejecting the GList's focus on modularity (and associated matters like the autonomy of syntax) as unnecessary.

Two problems arise here. First, it is unclear that non-modular cognitive architecture is sufficient to yield this behavior. For language use to be stimulus-free, some kind of internal self-prompting that independently accesses specifically linguistic resources appears necessary. There must be an independence of language from other cognitive systems of some minimal kind to enable this behavior. Even if UBL's cognitive architecture *were* sufficient to account for the possibility of "free" linguistic behavior, it appears substantially at odds with at least a minimal formulation of UG in GL.

Second, there is a deeper point here about the *acquisition problem* embedded in the Galilean Challenge. The child's acquisition of the ability to use language in ways that cannot be identified with relevant controlling stimuli, coupled with its frequent novelty and appropriateness, can be framed as a *poverty of stimulus* (PoS) problem. As Baker (2007) argued, the child must, in short order: *identify* that other individuals in their environment (e.g., parents) do not behave as automata based on these tripartite characteristics. Then, the child must learn that this behavior is accessible *to them*. Finally, the child must learn "*how* not to be an automaton – how to develop the capacity to use language in this way" (Baker, 2007, 241), at which point the investigation appears to hit a wall given that "we have no precise algorithmic way to specify the knowledge that this capacity depends on or the processes that it involves..." (Baker, 2007, 241). Framing acquisition in terms of the distribution (or quality) of data available to the child – though relevant in other contexts related to learnability – does not appear suited for the child's acquisition of this capacity.⁸

This point is worth elaboration. For the GList, the PoS argument "is no more than an invitation to look upon language acquisition as a process of biological development" (Collins, 2008, 102). It is striking to observe how the observations about ordinary language use embedded in the Galilean Challenge frame the child's development: communication with other humans is no doubt central to the child's cognitive development, though the child's learning appears simultaneously open-ended *yet* universally constrained. As Yiu et al. note generally, a human child is "curious, active, self-supervised, and intrinsically motivated" (Yiu et al., 2024, 880). Yet, in their open-ended explorations of their local environments, every

⁸ If one views Large Language Models (LLMs) as lending credence to UBL, they may argue that these models provide a way out of this acquisition problem. This is similar to Sripada et al.'s (2026) argument that LLMs can "illuminate" the "speaker's ability to freely select, from the vast array of grammatically possible sentences, utterances that are appropriate to the situation and linguistic context" (Sripada et al., 2026, 65). These brief arguments rest on a mistaken interpretation of Chomsky's use of the term *stimulus-freedom*. The authors appear to interpret this as a narrow break from stimulus-response psychology and language's generativity. However, stimulus-freedom is the ability to select an expression from an unbounded class of possible utterances that *is appropriate to a situation that does not cause it*. LLMs cannot lay claim to these particular characteristics (see, Carchidi, 2024). It is therefore questionable that LLMs could be used to properly frame this acquisition problem for human children.

child not only develops the capacity to pair form with meaning over an indefinite range of expressions,⁹ but *also* develops the ability to *use* this capacity in ways that are unfixed to local stimuli yet appropriate to circumstances. The child appears *constrained* to develop in a way that ensures language *can* be used “freely.” Indeed, language acquisition for the GList

seems to involve some specific kind of quasi-brute causal *process*...The relevant processes are not *cognitive*, but largely *brute* causal ones, implementing the principles largely without any rational assessment of them...Although a child may need to locally (dis)confirm the settings of the switches, she no more confirms the principles of UG than she confirms the principles by which her metabolic or immune system operates (Rey, 2020, 177-178) (emphasis in original).

The PoS refers to this “brute causal” process of implementing principles of UG rather than “some unusual probability assignments...” (Rey, 2020, 177).

Recall the core tenets of UBL: *meaning is use* and *structure emerges from use* (Tomasello, 2015, 89). Within a social context, carrying out a communicative function, grammatical structure emerges in the expression of meaning. This account of language acquisition frames the child as an *active participant* in their local environment during development, leveraging domain-general cognitive mechanisms – optimal to the task - in this process; “using language is a way of acting in the social world...” (Bunzeck et al., 2026, 35). The UBList sees Chomsky’s linguistics as anathema to this conception:

Chomsky (1957) argues forcefully against attempts to relate sentence structure to aspects of the environment, and Chomsky (1975) argues for a kind of autonomy of syntax that would cut it off from the pressures of communicative functions. In the Chomskian vision, language is pure and autonomous, unconstrained and unshaped by purpose or function (Bates & MacWhinney, 1989, 5).

Chomsky’s linguistics is viewed as a strange detachment from the *actual exercise of human language in communicative contexts*.

Should the UBList accept the Galilean Challenge’s depiction of ordinary language use, this would threaten to undermine UBL’s core tenets. The tenet that *structure emerges from use* implies that the co-mingling of knowledge and use of language during development precludes their separation in a science of language. The oddity here is that Chomsky’s view of human linguistic behavior appears to border on romantic, practically enthralled to a revised Cartesian conception of “creative” behavior in the relevant technical sense. It is precisely this view that *lends* to the segmentation of competence from performance and the “quasi-brute causal *process*” (Rey, 2020, 177) of language acquisition. The use of language develops in such a way as to exceed control by the external stimuli that so impress upon the usage-based approach to language (the communicative acts between parent and child during playtime, say). This does not mean that UBList has nothing to offer on such a depiction of linguistic behavior, but that a fairly rigid competence-performance distinction precludes UBL’s rejection of *a priori* boundaries between semantics and pragmatics. Similarly, the cognitive architecture that appears necessary to accommodate “free”

⁹ Even spontaneously developing communication systems with peculiarly specific grammatical structures, as in the case of second-generation Nicaraguan deaf children (see, Carruthers, 2026, 15-16).

language presents an acquisition challenge of no small measure, not evidently addressed via UBL's postulation of domain-general learning mechanisms.

The UBList and the GList appear to travel parallel to one another, seemingly prepared to meet at any moment. Somewhat tragically, it is unclear whether this meeting can happen without one diluting the core tenets of the other, at least in this context.

5.1.2 Idealization and Intractability

On (b), Chomsky and fellow travelers often invoke a “Galilean” style of explanation (the name overlapping with but somewhat incidental to the “Galilean Challenge”). As Nefdt points out, GLists differ from their opponents on “what counts as evidence and justification” in the study of language (Nefdt, 2025, 28). The Galilean style places significant emphasis on *abstraction* and *idealization*, with GLists analogizing the competence-performance distinction with “the elimination of friction and wind speed from Newton’s laws” (Nefdt, 2025, 29). In this way, GLists excise “extraneous” factors in biology from the study of an idealized system – the language faculty (Nefdt, 2025, 29).

Some UBLists have sharply critiqued Chomsky’s invocation of the Galilean style, illustrating a point of divergence between generative and usage-based philosophical commitments.¹⁰ Chater et al., for example, write:

Beyond skepticism, though, Galileo’s theorizing was based on two principles before all others... first, that we must look not to books but to Nature, the real phenomena, if we are to understand the world, and second, the language in which Nature is written is mathematical, which is to say, quantitative in character...it was the study of what things really do that allowed him to show that these patterns did not fit the received wisdom of the time, no matter how those theories satisfied the intellectual palates of established scholars...The fact is, there is no philosophy of science that allows one to ignore data (Chater et al., 2015, 96).

For Chater et al., the Galilean method concerns the explanation of phenomena *as they are observed* in the natural world. They also note, on the matter of what counts as *successful* explanatory theory:

Remember: Galileo wasn’t a success until his theories had been established empirically, both by matching prediction to observation and by showing that what had appeared to be false predictions were only apparent, and not real (Chater et al., 2015, 96).

The mark of success is *empirical* – when prediction *matches* observation. Idealization of the target of inquiry is not, on this reading, an appropriate condition on explanatory theory, at least to the extent emphasized by GLists. Transferred to linguistics, it is the *use* of language that is the proper object of explanation, rather than idealizations of the language faculty; it is precisely the factors GLists consider “extraneous” that UBLists consider central to explanation.

This is well-trodden ground. That said, it is intriguing that Chomsky’s first extended statement on the Galilean style in his 1980 *Rules and Representations* (R&R) explicitly analogizes between the Cartesian

¹⁰ It should be noted that, although some critics of Chomsky argue that he “coined a term for this unorthodox methodology” (Behme, 2014, 687), this is inaccurate. The term was coined instead by Husserl (1970/1936, 41), which Chomsky (1980, 8) explicitly acknowledged in the main text of *Rules and Representations*.

approach to “free” linguistic behavior and this style of explanation. The Galilean style, in *R&R*, serves as a “narrower version” of the Cartesian thesis that “free” behavior cannot be studied, though the *bases* underlying “free” action in human biology can be brought under the heel of natural science (Chomsky, 1980, 8). Drawing from physicist Steven Weinberg’s (1976) “The Forces of Nature,” Chomsky argues that there is “no present alternative to pursuing the “Galilean style” in the natural sciences at least” (Chomsky, 1980, 9). The idealization of the language faculty – the competence that remains with extraneous performance factors excised – follows from this limit on the scope of natural science.

Chomsky’s analogy between the Galilean style and the Cartesian thesis on “free” behavior is instructive, in ways potentially overlooked by UBLists: there is a class of problems for GLists – like ordinary language use – that appear to exceed the scope of natural science’s methods (at least currently). The UBList, were they to adopt the Galilean Challenge’s depiction of linguistic behavior, would again find themselves diluting their core tenets: abstracting *away* from ordinary language use and idealizing the object of inquiry (the cognitive capacities relevant to language). Chomsky’s analogy of the Cartesian thesis with the Galilean style should be kept in mind, as it follows the reasoning in our explication of GL above, yet is often presented separately.

6 Can UBL Meet the Galilean Challenge On Its Own Terms?

The UBList may take a different approach: the Galilean Challenge can be met on UBL’s own terms, apart from the contributions of GL.

Our task here is short-lived. As noted, UBL does not appear to provide the conceptual resources necessary to supplant GL’s invocation of a competence-performance distinction nor a means of circumventing the postulation of a modular and generative language faculty (even of some minimal kind). This turns principally on the intractability of “free” linguistic behavior, characterized according to its stimulus-freedom, unboundedness, and appropriateness to situations, as well as this ability’s acquisition. Although UBL provides a view of language acquisition in which the child is a critically active participant in their social context, the Galilean Challenge characterizes this behavior in a way that demands departure from UBL’s core tenets.

At this juncture, the UBList is more likely to suggest that any science of human language that excludes its ordinary use is simply ill-conceived. To this end, we turn to the likely way in which a UBList might show the Galilean Challenge to be ill-formulated (and thus of little or no concern).

7 Is the Galilean Challenge Ill-Formulated?

The UBList may argue that the Galilean Challenge is ill-formulated. Specifically, the UBList may argue that Chomsky smuggles a *GL-specific* assumption about the nature of linguistic *novelty* into his presentation of the Galilean Challenge.

The Galilean Challenge is to explain how it is that humans can select from an unbounded class of possible expressions an utterance that is *appropriate* to a situation but not *caused* by the situation. Part of the Galilean Challenge, then, assumes that ordinary language use is *sufficiently novel* and capable of continuous novelty over an unbounded range, such that new expressions are produced and appropriately selected by individuals for new situations. The UBList may argue against this assumption. UBLists have long drawn attention to the “importance of similarity to prior utterances or parts of utterances in the

production of novel utterances” (Bybee, 2010, 59). Bybee argues that “it is important to point out that many utterances are not novel or at least contain parts that are not thoroughly novel” (Bybee, 2010, 59). She continues:

In the context of assertions about the infinite creative capacity of the human language ability and the apparent fact that humans can express any concept through language, the fact that our actual utterances contain many pre-packaged word sequences comes as a bit of a surprise. Corpus studies...have shown that even on fairly conservative counts, about 55 per cent of word choices are not ‘free’ but rather are determined by a larger chunk, or prefab (prefabricated unit) (Bybee, 2010, 59-60).

This is not to diminish the *capacity* for linguistic novelty, as Bybee notes this capacity “is what endows language with its much-lauded creativity” (Bybee, 2010, 60). The noted “importance” of similarity to prior utterances is part-and-parcel of UBL’s attempt to *explain* novelty via the *actual patterns of linguistic behavior*, including the frequency of specific constructions.

Plausibly, GLists like Chomsky exaggerate the *frequency of novelty* within ordinary language use. Indeed, there is at least tentative empirical evidence of duplication of sentences in English text corpora, although its preponderance is mostly low, if higher in speech-recorded corpora (Ring, 2025). This exaggeration, on the UBList’s critique, is embedded in Chomsky’s presentation of the Galilean Challenge: GL is *framed* in relation to the infinite generativity of human language, conceived *independently* of its use, making the Galilean Challenge a reflection of generative assumptions. UBL explicitly rejects this framing.

7.1 Reduced, Actual Novelty Does Not Undermine the Galilean Challenge

Although the UBList is correct to critique the GList’s tendency to exaggerate the novelty of ordinary utterances, it is unclear that this undermines the Galilean Challenge’s presentation within GL. The challenge of explaining how individuals speak in novel ways that appear unfixed to circumstances yet are appropriate to circumstances remains intact. Indeed, the characterization of language as *generative* is not rejected by UBL – where UBL diverges from GL is how to characterize this capacity for generativity within a theoretical framework. It is here that UBL rejects abstractly conceived notions of grammar in isolation from use. This is distinct from rejecting the notion that individuals are in fact capable of producing and understanding an indefinite range of linguistic expressions.

Chomsky’s presentation of the Galilean Challenge *does* frame the problem in terms familiar to GL, idealizing the capacity for linguistic novelty. This point overlaps with the notion of *recursion*. This matter will not detain us, though Huybregts’ remarks are worth noting:

We should be careful, however, not to confuse the *infinite productivity* of the generative procedure (a function in intension) with the *cardinality* of the decidable set of structures it generates (the well-formed formulas defined in extension)...Without contradiction, finite outputs can be generated by recursive systems (Huybregts, 2019, 3) (emphasis in original).

Although human language provides the capacity for infinite output, “linguistic output (“performance”) must necessarily be strictly finite for other, biophysical reasons, quite extraneous to the generative procedure itself” (Huybregts, 2019, 4).

With this in mind, we note that, although UBL adopts a different approach to the study of the capacity for novelty, this alternative approach does not provide the resources to undermine the problem posed by the Galilean Challenge. Instead, it may be more accurate to frame the novelty component of the Galilean Challenge as the problem of human language use that is unfixed yet appropriate to circumstances, *on the assumption that humans can and will produce new expressions*. This is an eminently reasonable assumption.

8 Discussion

The Galilean Challenge of explaining how individuals select from an unbounded range of possible expressions an utterance that is appropriate to a situation but not caused by the situation – characterized according to ordinary language use’s *stimulus-freedom*, *unboundedness*, and *appropriateness to situations* – has chiefly been articulated by Chomsky as the organizing problem of linguistics from his preferred generative framework. This paper posed the Galilean Challenge to UBL. Through an evaluation of three plausible responses by UBL, it was found that UBL cannot meet the Challenge without diluting its core tenets.

In short order, we found the following: (1) UBL and GL are fundamentally incompatible in their attempts to meet the Galilean Challenge, with the adjustments required of UBL too severe to maintain the integrity of the approach’s core tenets; (2) UBL, largely in virtue of its incompatibility with GL, is unable to meet the Galilean Challenge on its own terms; and (3) UBL is unable to show, with its own approach-specific resources, that the Galilean Challenge is ill-formulated.

To be sure, the foregoing arguments have limitations. For one thing, the Galilean Challenge was posed to UBL without an extended and broader investigation of its merits, providing only one UBL-specific argument against its generative presentation. Indeed, it could turn out that UBL does not *need* to refute or meet the Galilean Challenge, as the approach was never *designed* for this problem. That said, this risks evading the problem: the *question* is whether UBL can meet the Galilean Challenge. That it was not designed for this purpose could merely be a fundamental shortcoming of this approach to a science of language. Nor is Chomsky’s articulation of the Galilean Challenge obviously unreasonable. Even if one does not wish to become mired in debates about free will, human language crucially *does* appear available for use across such an indefinite range of situations and purposes as to warrant intense analysis. The difference between a capacity that enables unbounded linguistic productivity yet is fixed to stimuli, and one that enables it but can be recruited without reliance on fixed sets of stimuli, is a difference in *kind*, not degree. This evidently has not changed since Chomsky’s critique of Skinner.

One could also argue that this paper’s explication of both UBL and GL are unnecessarily rigid and unduly constrain the conceptual reaches of both, particularly with respect to the competence-performance distinction and the (non)modularity of language. A more flexible view meshes with Rastelli’s (2025) argument in favor of third-way linguistics, echoed by Ring (2025, 4-5), in which the values of both approaches should be identified and selectively extracted as found. That said, if there is merit to the Galilean Challenge, one should be cautious about forced holism. When the two approaches are viewed in relation to this Challenge, their incompatibilities appear severe, rather stringent, and important.

The conclusions of this paper are nevertheless somewhat tragic. It is unclear, even if the Galilean Challenge does have merit, that GL and UBL should remain hopelessly at odds. The nearly romantic depiction of human language use offered by Chomsky *seems* to be a fellow traveler of UBL’s position that

the child's active participation in their environment during language acquisition is the locus of investigation.

A closing line of thought therefore looks to a recent debate that has arisen between specific factions within UBL and GL over the role(s) of Large Language Models (LLMs) in linguistics. Futrell & Mahowald (F&M) (2025) argue that LLMs should play a significant role in linguistics for mutual benefit. In a commentary, Bunzeck et al. (2026) argue that, although LLMs are *consistent* with UBL, their use in UBL is suspect because they “entirely lack the socio-cognitive grounding that usage-based linguistics considers constitutive of language” (Bunzeck et al., 2026, 35). They write: “In fact, even the very notion of language (use) is fundamentally different between language models and humans...In usage-based linguistics, using language is a way of acting in the social world...” (Bunzeck et al., 2026, 35). In response, F&M agree that there “is something off about merely learning from co-occurrence statistics, without having rich communicative goals” but argue this limitation can be overcome with “the right training regime” (F&M, 2026, 77). They reference behavioral studies of artificial intelligence agents placed “in much richer settings” and, specifically, “a setting that requires them to have goals (e.g., helping a user write successful code or complete a task)” (F&M, 2026, 78).

GL, through the Galilean Challenge, has preempted this concern about LLMs. It does so in two important ways. First, by recognizing that “Language is a process of free creation; its laws and principles are fixed, *but the manner in which the principles of generation are used is free and infinitely varied*” (Chomsky, 1999, 99) (emphasis added). The relation of a human's use of language to their environment appears *qualitatively different* from the relation of an LLM's outputs to its environment, the latter fixed to external stimuli (e.g., prompts). Even if one could find a productive way to frame ordinary human linguistic behavior in causal terms (that does not rely on post-hoc attribution and the like), something qualitatively different would remain between a human's relation to their environment and an LLM's. There appears to be no limit to the number or kinds of situations or purposes for which humans can use language nor any obvious barrier, save for contrived situations like coercive environments, to an individual not exercising their linguistic capacity despite the availability of what, in other contexts, may be considered controlling stimuli. They are simply different.

Second, it sheds light on the deficiency of F&M's proposed resolution: instilling goals from the outside in AI agents – however rich the settings may be – is a crude sense of the notion *communication*. It is, more importantly, a highly contrived notion of language *use*. Human language use – by virtue of its stimulus-freedom, unboundedness, yet appropriateness to situations – is not a functional activity. The LLM (in either chatbot or agentic form) is closer to the aforementioned coerced behavior than ordinary language use in that its outputs or actions are necessary functions of the local environment.

That the Galilean Challenge appears to preempt the concern of some UBL factions over the use of LLMs in linguistics should give one pause. The foregoing arguments and their implications are not, to be sure, undermined by this – GL and UBL still appear fundamentally at odds. Yet, there is an opportunity that presents itself in this context for an alliance between specific factions within UBL and GL broadly – even if temporary – on their suspicion of LLMs in linguistics driven by a mutual respect for the distinctiveness of ordinary human language use.

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